

Shubha Tanaya

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Robotics System Development (MRSD) | GPA: 4

May 2027

Relevant Courses: Robot Mobility; SLAM; Robot Autonomy; Manipulation Estimation and Control; Computer Vision

Indian Institute of Technology (IIT)

Ropar, India

Bachelor of Technology in Mechanical Engineering | GPA: 8.31/10.0

May 2025

Completed Relevant Coursework: Robotic Manipulators, Deep Learning of Physical Systems, Multibody Dynamics

SKILLS

Programming & Middleware: C++, Python, MATLAB, ROS2 (Humble), Micro XRCE-DDS, Linux (Ubuntu), Git, Docker

Framework & Software: PX4, Gazebo, TensorFlow, SolidWorks, MuJoCo, MATLAB/Simulink, Nav2, Cartographer SLAM

Platforms: NVIDIA Jetson Orin NX, Arduino, Raspberry Pi, Linux

PROJECTS

F1Tenth Autonomous Racing

DRIVE Lab

Carnegie Mellon University

- Delivered a high-speed autonomous racing demonstration at the Safety21 Consortium, optimizing ROS2 software stack for robust, real-time performance on NVIDIA Jetson Orin NX embedded hardware.
- Reduced localization drift in dynamic, unstructured environments by integrating LiDAR-based SLAM with a Particle Filter, enabling high-frequency pose estimation and highly precise vehicle positioning under racing conditions.
- Maximized racing efficiency and safety through a dual-tier navigation system, combining Reactive Follow-the-Gap for dynamic obstacle avoidance and Pure Pursuit for high-speed trajectory tracking and optimal path planning.

UGV-UAV Collaboration for Exploratory Mapping

Indian Institute of Technology

Modular and Intelligent Robotics Lab

Aug 2024-May 2025

- Developed high-fidelity Digital Twins for heterogeneous robots using ROS2 Humble and PX4/Gazebo, synchronizing UGV/UAV dynamics via Micro XRCE-DDS to streamline multi-agent coordination.
- Architected an autonomous navigation stack integrating Cartographer SLAM for UGV mapping and Nav2 Planner for dynamic obstacle avoidance, alongside PID-based waypoint following for UAV precision landing.
- Executed hardware-in-the-loop (HIL) validation on Turtlebot3 and Holybro X500 V2, integrating LiDAR and RGB-D sensor fusion to achieve real-time 3D occupancy grid mapping in unknown environments.

Automated Subsurface Target Detection & GPR Signal Processing

Indian Institute of Technology

Development Engineering Project

- Engineered a synthetic GPR dataset by modeling heterogeneous soil dielectric properties using GPRMax (FDTD-based); simulated 5,000+ A-scan and B-scan profiles to replicate diverse subsurface clutter and variable target burial depths.
 - Architected a computer vision pipeline in OpenCV (Otsu thresholding, Hough Transform) and fine-tuned YOLOv5 for Automated Target Recognition, significantly improving classification mAP over traditional signal processing baselines.
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EXPERIENCE

BMW Study, Carnegie Mellon University

Nov 2025 – present

Industry-Sponsored Project

- Robotics Applications in Automotive: Recommended technology adoption strategies using Generative Engineering and CAD/CAE-enabled Digital Twins within the Industrial Metaverse to improve system reliability and validation.

AIR Lab, Carnegie Mellon University

Oct 2025 – present

Research Assistant

- Aerial Manipulation Project: Implemented whole-body Inverse Kinematics for a UAV-manipulator using CasADi, DH forward kinematics, and SO(3) quaternion orientation error, with migration toward cuSaDi.

Collins Aerospace- Global Engineering and Technology Centre

Bengaluru, India

Project Intern

June 2024 - August 2024

- Optimized nacelle structural design workflows by developing a data-driven ML model for fan cowl parametrization, accurately predicting pressure-thickness for RFQ/RFP, cutting estimation time by 75% and enhancing scalability.
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LEADERSHIP

Secretary, Society of Mechanical Engineers- Indian Institute of Technology, India

Aug 2024-May 2025

Mentor, Secretary, Representative, Automotive Club- Indian Institute of Technology, India

Aug 2022-May 2025